CORE PURE / DIFFERENTIAL EQUATIONS

# 2nd Order Differential Equations

Mark scheme 12 questions · 125 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

## Questions


| Question | Marks by part | Marks | Page |
|---|---|---|---|
| Question 1 | (a) 2 (b) 4 | 6 | 4 |
| Question 2 | | 11 | 5 |
| Question 3 | (a) 7 (b) 4 | 11 | 6 |
| Question 4 | | 11 | 7 |
| Question 5 | (a) 2 (b) 5 | 7 | 8 |
| Question 6 | | 11 | 9 |
| Question 7 | | 11 | 10 |
| Question 8 | | 11 | 11 |
| Question 9 | (a) 2 (b) 10 | 12 | 12 |
| Question 10 | (a) 2 (b) 10 | 12 | 13 |
| Question 11 | (a) 2 (b) 10 | 12 | 14 |
| Question 12 | | 10 | 15 |
| | **Total** | **125** | |

# Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

### Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

| Code | Meaning |
|---|---|
| **M1** | Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted. |
| **dM1** | Dependent method: as M1, but only if the method mark before it was earned, unless its note names another. |
| **A1** | Accuracy: a correct answer or step. It needs the method mark it follows. |
| **A1ft** | As A1, following through from an earlier answer that the note names. |
| **A1 AG** | The last mark of a part whose answer is printed in the question: complete working, with no errors. |
| **B1** | A correct value, fact or statement. It needs no method mark. |
| **B1ft** | As B1, following through from an earlier answer that the note names. |
| **E1** | A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark. |

### Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

| | | | |
|---|---|---|---|
| **ft** | follow through<br>Correct work from the student's own earlier wrong value earns the mark. | **AG** | answer given<br>The answer is printed in the question, so every step to it must be shown. |
| **cao** | correct answer only<br>No follow-through from an earlier error. | **cso** | correct solution only<br>No error anywhere in the part, not just in its answer. |
| **oe** | or equivalent | **awrt** | answers which round to |
| **soi** | seen or implied<br>Earned if written, or if later working shows it was used. | **isw** | ignore subsequent working<br>A correct answer, once reached, isn't spoiled by later working. |
| **SC** | special case<br>Credit, as the note says, for work outside the main method. | **their** | the student's own earlier value |
| **condone** | a slip that doesn't cost the mark | [ ] | working the mark doesn't need |
| **s.f.** | significant figures | **d.p.** | decimal places |

**Reading the scheme**

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

**Accuracy and answers**

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

**Value of $g$**

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state $g$, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **1(a)** | $\lambda^2 + 4 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by any method. |
| | $\lambda = \pm 2i$<br>$(y =)\ C_1 \cos 2x + C_2 \sin 2x$ | **A1** | Must be in terms of $\cos 2x$ and $\sin 2x$, oe such as $R\cos(2x + \alpha)$. |
| | | **[2]** | |
| **1(b)** | $y = x(A\cos 2x + B\sin 2x)$<br>$y' = A\cos 2x + B\sin 2x$<br>$\quad + x(-2A\sin 2x + 2B\cos 2x)$<br>$y'' = -4A\sin 2x + 4B\cos 2x$<br>$\quad - 4x(A\cos 2x + B\sin 2x)$<br>$4B\cos 2x - 4A\sin 2x = 8\cos 2x - 12\sin 2x$<br>$\Rightarrow A = \ldots,\ B = \ldots$ | **M1** | Differentiates the given form twice by the product rule, substitutes into the equation and equates coefficients of $\cos 2x$ and $\sin 2x$. Condone slips in differentiation. |
| | $A = 3$ or $B = 2$ | **A1** | |
| | $x(3\cos 2x + 2\sin 2x)$ | **A1** | Both constants correct, in the particular integral or stated. |
| | $(y =)\ C_1 \cos 2x + C_2 \sin 2x$<br>$\quad + x(3\cos 2x + 2\sin 2x)$ | **A1ft** | ft their complementary function from (a) plus their particular integral of the form $x(p\cos 2x + q\sin 2x)$ with numerical $p$, $q$. |
| | | **[4]** | |
| | **Total for Question 1** | **6** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **2** | $\lambda^2 + 4\lambda + 5 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by formula, completing the square or calculator. |
| | $\lambda = -2 \pm i$<br>$(y =)\ e^{-2t}(A\cos t + B\sin t)$ | **A1** | |
| | $y = a\cos t + b\sin t$ | **B1** | |
| | $\frac{\mathrm{d}y}{\mathrm{d}t} = -a\sin t + b\cos t,\ \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -a\cos t - b\sin t$<br>$(-a + 4b + 5a)\cos t + (-b - 4a + 5b)\sin t$<br>$= 8\cos t$ | **M1** | Differentiates their trial function twice, substitutes into the equation and equates coefficients of $\cos t$ and $\sin t$. Condone sign slips in the derivatives. |
| | $4a + 4b = 8, \quad -4a + 4b = 0$ | **A1** | oe |
| | $\Rightarrow a = \ldots,\ b = \ldots$ | **dM1** | Solves their two linear equations for both $a$ and $b$, by any means. |
| | $(y =)\ \cos t + \sin t$ | **A1** | |
| | $y = e^{-2t}(A\cos t + B\sin t) + \cos t + \sin t$ | **A1ft** | ft their complementary function with two arbitrary constants plus their particular integral; must be $y = \ldots$ in terms of $t$. |
| | $t = 0,\ y = 3:\ 3 = A + 1 \Rightarrow A = \ldots$ | **M1** | Substitutes $t = 0$, $y = 3$ into their general solution with two arbitrary constants. |
| | $\frac{\mathrm{d}y}{\mathrm{d}t} = -2e^{-2t}(A\cos t + B\sin t)$<br>$+ e^{-2t}(-A\sin t + B\cos t) - \sin t + \cos t$<br>$t = 0:\ 1 = -2A + B + 1 \Rightarrow B = \ldots$ | **M1** | Differentiates their general solution, using the product rule on the exponential term, substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 1$ and solves for both constants. Condone sign slips. |
| | $y = e^{-2t}(2\cos t + 4\sin t) + \cos t + \sin t$ | **A1** | Must be stated as $y = \ldots$ in terms of $t$; $A = 2$, $B = 4$ alone is not enough. |
| | **Total for Question 2** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **3(a)** | $\lambda^2 + 6\lambda + 13 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by formula, completing the square or calculator, reaching complex roots. |
| | $\lambda = -3 \pm 2i$<br>$(y =)\, e^{-3t}(A\cos 2t + B\sin 2t)$ | **A1** | Correct complementary function oe; roots alone are not enough. |
| | $y = a\sin 2t + b\cos 2t$ | **B1** | |
| | $-4a\sin 2t - 4b\cos 2t + 6(2a\cos 2t - 2b\sin 2t)$<br>$\quad + 13(a\sin 2t + b\cos 2t) = 75\sin 2t$<br>Equates coefficients of $\sin 2t$ and $\cos 2t$ | **M1** | Differentiates their trial function twice, substitutes into the equation and equates coefficients of $\sin 2t$ and $\cos 2t$. Condone sign and coefficient slips in the derivatives. |
| | $9a - 12b = 75, \quad 12a + 9b = 0$ | **A1** | oe |
| | $a = 3,\ b = -4$<br>$(y =)\, 3\sin 2t - 4\cos 2t$ | **A1** | |
| | $y = e^{-3t}(A\cos 2t + B\sin 2t) + 3\sin 2t - 4\cos 2t$ | **A1ft** | ft their complementary function with two arbitrary constants plus their particular integral with none. Needs $y =$. |
| | | **[7]** | |
| **3(b)** | $t \to \infty \Rightarrow e^{-3t}(A\cos 2t + B\sin 2t) \to 0$<br>for all $A, B$, so $y \approx 3\sin 2t - 4\cos 2t$ | **E1** | States that $e^{-3t} \to 0$, so the complementary function vanishes whatever $A$ and $B$ (or the initial conditions), leaving the particular integral. |
| | $R\sin(2t - \phi) = R\cos\phi\sin 2t - R\sin\phi\cos 2t$<br>$R\cos\phi = 3, \quad R\sin\phi = 4$ | **M1** | Expands $R\sin(2t - \phi)$ and equates coefficients with their particular integral of the form $p\sin 2t + q\cos 2t$, or uses $R = \sqrt{p^2 + q^2}$ and $\tan\phi$ with their $p, q$. Condone sign slips. |
| | $R = 5$ | **A1** | |
| | $\phi = \arctan\frac{4}{3}$ $[\approx 0.927]$ | **A1** | $\arctan\frac{4}{3}$ oe, such as $\arcsin\frac{4}{5}$, or awrt 0.927. Not in degrees. |
| | | **[4]** | |
| | **Total for Question 3** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **4** | $\lambda^2 + 2\lambda + 5 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by formula, completing the square or calculator, leading to $\lambda = \ldots$. |
| | $\lambda = -1 \pm 2i$<br>$(y =)\ e^{-t}(A\cos 2t + B\sin 2t)$ | **A1** | Any equivalent real form, such as $Re^{-t}\cos(2t + \alpha)$. |
| | $(y =)\ a\cos t + b\sin t$ | **B1** | |
| | $(-a\cos t - b\sin t) + 2(-a\sin t + b\cos t)$<br>$+ 5(a\cos t + b\sin t) = 10\cos t$ | **M1** | Differentiates their trial function twice, substitutes into the equation and equates coefficients of $\cos t$ and $\sin t$. Condone sign slips in the derivatives. |
| | $4a + 2b = 10, \quad 4b - 2a = 0$ | **A1** | oe |
| | $\Rightarrow a = \ldots,\ b = \ldots$ | **dM1** | Solves their two simultaneous equations for $a$ and $b$ by any means. |
| | $a = 2,\ b = 1$<br>$(y =)\ 2\cos t + \sin t$ | **A1** | |
| | $y = e^{-t}(A\cos 2t + B\sin 2t) + 2\cos t + \sin t$ | **A1ft** | ft their complementary function, with two arbitrary constants, plus their particular integral. Needs $y =$. |
| | $5 = A + 2 \Rightarrow A = \ldots\ [= 3]$ | **M1** | Substitutes $t = 0$, $y = 5$ into their general solution with two arbitrary constants to find one constant. |
| | $\frac{\mathrm{d}y}{\mathrm{d}t} = -e^{-t}(A\cos 2t + B\sin 2t)$<br>$+ e^{-t}(-2A\sin 2t + 2B\cos 2t)$<br>$- 2\sin t + \cos t$<br>$2 = -\text{“3”} + 2B + 1 \Rightarrow B = \ldots\ [= 2]$ | **M1** | Differentiates their general solution, using the product rule on any product, substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$ and solves for the other constant. Condone sign slips. |
| | $y = e^{-t}(3\cos 2t + 2\sin 2t) + 2\cos t + \sin t$ | **A1** | cso. Needs $y =$. |
| | **Total for Question 4** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **5(a)** | $\lambda^2 + 4\lambda + 13 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by formula, completing the square or calculator, reaching $\lambda = \ldots$. |
| | $\lambda = -2 \pm 3i$<br>$y = e^{-2x}(A\cos 3x + B\sin 3x)$ | **A1** | Real form, oe. |
| | | **[2]** | |
| **5(b)** | $y = e^{-2x}(a\cos x + b\sin x)$ | **B1** | |
| | $\frac{\mathrm{d}y}{\mathrm{d}x} = e^{-2x}[(b-2a)\cos x - (a+2b)\sin x]$<br>$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = e^{-2x}[(3a-4b)\cos x + (4a+3b)\sin x]$<br>$\Rightarrow 8a\cos x + 8b\sin x = 6\cos x - 8\sin x$<br>$\Rightarrow a = \ldots,\ b = \ldots$ | **M1** | Differentiates their trial form twice using the product rule, substitutes into the equation and equates coefficients of $\cos x$ and $\sin x$. Condone sign and coefficient slips. |
| | $a = \frac{3}{4}$ or $b = -1$ | **A1** | Either constant correct. |
| | $(y =)\ e^{-2x}\left(\frac{3}{4}\cos x - \sin x\right)$ | **A1** | |
| | $y = e^{-2x}\left(A\cos 3x + B\sin 3x + \frac{3}{4}\cos x - \sin x\right)$ | **A1ft** | ft their complementary function from (a), with two arbitrary constants, plus their particular integral; must start $y =$. |
| | | **[5]** | |
| **Total for Question 5** | | **7** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **6** | $\lambda^2 + 4\lambda + 13 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by formula with their values, completing the square or calculator, leading to $\lambda = \ldots$ |
| | $\lambda = -2 \pm 3i$<br>$y = e^{-2t}(A\cos 3t + B\sin 3t)$ | **A1** | Any equivalent real form; the roots alone are not enough. |
| | $(y =)\ a\cos 3t + b\sin 3t$ | **B1** | |
| | $\frac{\mathrm{d}y}{\mathrm{d}t} = -3a\sin 3t + 3b\cos 3t$<br>$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -9a\cos 3t - 9b\sin 3t$<br>$(-9a + 12b + 13a)\cos 3t$<br>$+ (-9b - 12a + 13b)\sin 3t$<br>$= 28\cos 3t - 4\sin 3t$ | **M1** | Differentiates their trial function twice, substitutes all three terms and equates coefficients of $\cos 3t$ and $\sin 3t$. Condone sign and coefficient slips. |
| | $4a + 12b = 28,\ 4b - 12a = -4$ | **A1** | oe |
| | $a = \ldots,\ b = \ldots$ | **dM1** | Solves their two equations simultaneously by any means, calculator included, to find both $a$ and $b$. |
| | $y = \cos 3t + 2\sin 3t$ | **A1** | |
| | $y = e^{-2t}(A\cos 3t + B\sin 3t) + \cos 3t + 2\sin 3t$ | **A1ft** | ft their complementary function, with two arbitrary constants, plus their particular integral with no arbitrary constants. |
| | $5 = A + 1\ [\Rightarrow A = 4]$ | **M1** | Substitutes $t = 0$, $y = 5$ into their general solution, with two arbitrary constants, to form an equation in the constants. |
| | $\frac{\mathrm{d}y}{\mathrm{d}t} = -2e^{-2t}(A\cos 3t + B\sin 3t)$<br>$+ e^{-2t}(3B\cos 3t - 3A\sin 3t)$<br>$- 3\sin 3t + 6\cos 3t$<br>$7 = -2A + 3B + 6 \Rightarrow B = \ldots$ | **M1** | Differentiates their general solution, with the product rule on the complementary function, substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 7$ and solves for both constants. Condone sign slips. |
| | $y = e^{-2t}(4\cos 3t + 3\sin 3t) + \cos 3t + 2\sin 3t$ | **A1** | Needs $y = \ldots$; $A = 4$, $B = 3$ alone is not enough. |
| | **Total for Question 6** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **7** | $\lambda^2 - \lambda - 6 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it: factors whose constants multiply to their constant term, the formula, completing the square or a calculator. |
| | $\lambda = 3,\ -2$<br>$(y =)\ Ae^{3x} + Be^{-2x}$ | **A1** | |
| | $(y =)\ a\sin x + b\cos x$ | **B1** | |
| | $y' = a\cos x - b\sin x,\ y'' = -a\sin x - b\cos x$<br>$(-a\sin x - b\cos x) - (a\cos x - b\sin x)$<br>$-6(a\sin x + b\cos x) = 50\sin x$ | **M1** | Differentiates a particular integral of the form $a\sin x + b\cos x$ twice, substitutes into the equation and equates coefficients of $\sin x$ and $\cos x$. Condone sign slips. |
| | $b - 7a = 50,\quad -a - 7b = 0$ | **A1** | oe |
| | $\Rightarrow a = \ldots,\ b = \ldots$ | **dM1** | Solves their two simultaneous equations for $a$ and $b$, by any means including a calculator. |
| | $(y =)\ -7\sin x + \cos x$ | **A1** | |
| | $y = Ae^{3x} + Be^{-2x} - 7\sin x + \cos x$ | **A1ft** | ft their complementary function plus their particular integral; must be $y = \ldots$ in terms of $x$. |
| | $3 = A + B + 1$ | **M1** | Substitutes $x = 0$, $y = 3$ into their general solution, of the form complementary function with two arbitrary constants plus particular integral. |
| | $y' = 3Ae^{3x} - 2Be^{-2x} - 7\cos x - \sin x$<br>$-6 = 3A - 2B - 7 \Rightarrow A = \ldots,\ B = \ldots$ | **M1** | Differentiates their general solution, substitutes $x = 0$, $y' = -6$, and solves with their first equation for $A$ and $B$. Condone slips in differentiation. |
| | $y = e^{3x} + e^{-2x} - 7\sin x + \cos x$ | **A1** | Must be $y = \ldots$. |
| | **Total for Question 7** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **8** | $\lambda^2 + 4\lambda + 13 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by the formula with their values, completing the square or calculator. |
| | $\lambda = -2 \pm 3i$<br>$(y =)\ e^{-2t}(A\cos 3t + B\sin 3t)$ | **A1** | |
| | $y = e^{-2t}(a\sin t + b\cos t)$ | **B1** | |
| | $\frac{\mathrm{d}y}{\mathrm{d}t} = -2e^{-2t}(a\sin t + b\cos t)$<br>$\quad + e^{-2t}(a\cos t - b\sin t)$<br>$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \ldots$ | **M1** | Differentiates their particular integral twice, using the product rule on $e^{-2t}$ times the trigonometric terms. Condone sign and coefficient slips. |
| | $\frac{\mathrm{d}y}{\mathrm{d}t} = e^{-2t}[(-2a - b)\sin t + (a - 2b)\cos t]$<br>$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = e^{-2t}[(3a + 4b)\sin t + (3b - 4a)\cos t]$ | **A1** | Both derivatives correct; need not be simplified. |
| | $e^{-2t}(8a\sin t + 8b\cos t) = 8e^{-2t}\sin t$<br>$\Rightarrow a = \ldots,\ b = \ldots$ | **dM1** | Substitutes their derivatives into the equation and equates coefficients of $\sin t$ and $\cos t$ to find $a$ and $b$. Condone slips. |
| | $(y =)\ e^{-2t}\sin t$ | **A1** | |
| | $y = e^{-2t}(A\cos 3t + B\sin 3t + \sin t)$ | **A1ft** | ft their complementary function with two arbitrary constants plus their particular integral, which has none. Needs $y =$. |
| | $t = 0,\ y = 1 \Rightarrow A = \ldots\ [= 1]$ | **M1** | Substitutes $t = 0$, $y = 1$ into their general solution, complementary function plus particular integral, to find $A$. |
| | $\frac{\mathrm{d}y}{\mathrm{d}t} = -2e^{-2t}(A\cos 3t + B\sin 3t + \sin t)$<br>$\quad + e^{-2t}(-3A\sin 3t + 3B\cos 3t + \cos t)$<br>$2 = -2\text{“}A\text{”} + 3B + 1 \Rightarrow B = \ldots\ [= 1]$ | **M1** | Differentiates their general solution using the product rule, substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$ and solves for $B$ with their $A$. Condone sign slips. |
| | $y = e^{-2t}(\cos 3t + \sin 3t + \sin t)$ | **A1** | cao, with $y =$. |
| | **Total for Question 8** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **9(a)** | $(1+\tanh x)\cosh x = \cosh x + \sinh x$<br>$= \frac{e^x+e^{-x}}{2} + \frac{e^x-e^{-x}}{2}$ | **M1** | Uses $\tanh x = \frac{\sinh x}{\cosh x}$ and exponential definitions, or writes $\tanh x$ and $\cosh x$ in exponentials directly. Condone a sign slip. |
| | $= e^x$ | **A1 AG** | Correct exponential expression leading without error to $e^x$; combined steps accepted. |
| | | **[2]** | |
| **9(b)** | $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 3e^x$ | **B1** | |
| | $\lambda^2 - 2\lambda + 1 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by any means, leading to $\lambda = \ldots$. |
| | $(y =)(A + Bx)e^x$ | **A1** | |
| | $y = ax^2e^x$ | **B1** | Or any form containing an $x^2e^x$ term, such as $(ax^2 + bx + c)e^x$. |
| | $y' = e^x(ax^2 + 2ax)$, $y'' = e^x(ax^2 + 4ax + 2a)$<br>$\Rightarrow 2ae^x = 3e^x \Rightarrow a = \ldots$ | **M1** | Differentiates their particular integral twice, substitutes into the differential equation with their right-hand side and solves for $a$. Condone slips in differentiation. |
| | $a = \frac{3}{2}$ | **A1** | Or $y = \frac{3}{2}x^2e^x$. |
| | $y = (A + Bx)e^x + \frac{3}{2}x^2e^x$ | **A1ft** | ft their complementary function plus their particular integral, with two arbitrary constants. |
| | $x = 0,\ y = 1 \Rightarrow A = \ldots\ [= 1]$ | **M1** | Substitutes $x = 0$, $y = 1$ into their general solution containing two arbitrary constants. |
| | $\frac{\mathrm{d}y}{\mathrm{d}x} = e^x\left(A + Bx + \frac{3}{2}x^2\right) + e^x(B + 3x)$<br>$\Rightarrow 0 = A + B \Rightarrow B = \ldots$ | **M1** | Differentiates their general solution, product rule on at least one term, substitutes $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ and solves for the constants. |
| | $y = e^x\left(\frac{3}{2}x^2 - x + 1\right)$ | **A1** | oe, as $y$ in terms of $x$. |
| | | **[10]** | |
| | **Total for Question 9** | **12** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **10(a)** | $\frac{\frac{1}{2}(e^x+e^{-x})-\frac{1}{2}(e^x-e^{-x})}{\frac{1}{2}(e^x+e^{-x})+\frac{1}{2}(e^x-e^{-x})}$ | **M1** | Replaces $\cosh x$ and $\sinh x$ by their exponential definitions in both numerator and denominator. |
| | $=\dfrac{e^{-x}}{e^x}=e^{-2x}$ | **A1 AG** | Correct substituted expression simplified without error to the printed result; combined steps accepted. |
| | | **[2]** | |
| **10(b)** | $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+4\frac{\mathrm{d}y}{\mathrm{d}x}+4y=4e^{-2x}$ | **B1** | May be implied by working with right-hand side $4e^{-2x}$. |
| | $\lambda^2+4\lambda+4=0\Rightarrow\lambda=\ldots$ | **M1** | Forms the auxiliary equation and solves it by any method. |
| | $(y=)(A+Bx)e^{-2x}$ | **A1** | |
| | $(y=)\,ax^2e^{-2x}$ | **B1** | Or $(ax^2+bx+c)e^{-2x}$. |
| | $\frac{\mathrm{d}y}{\mathrm{d}x}=(2ax-2ax^2)e^{-2x}$<br>$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=(2a-8ax+4ax^2)e^{-2x}$<br>$\Rightarrow 2ae^{-2x}=4e^{-2x}\Rightarrow a=\ldots$ | **M1** | Differentiates their particular integral of the form $ax^2e^{-2x}$ twice by the product rule, substitutes into the equation and solves for $a$. Condone coefficient slips. |
| | $(y=)\,2x^2e^{-2x}$ | **A1** | |
| | $y=(A+Bx)e^{-2x}+2x^2e^{-2x}$ | **A1ft** | ft their complementary function with two arbitrary constants plus their non-zero particular integral. |
| | $x=0,\ y=1\Rightarrow A=\ldots\ [=1]$ | **M1** | Substitutes $x=0$, $y=1$ into their general solution with two arbitrary constants. |
| | $\frac{\mathrm{d}y}{\mathrm{d}x}=-2e^{-2x}(2x^2+Bx+A)+e^{-2x}(4x+B)$<br>$1=B-2A\Rightarrow B=\ldots$ | **M1** | Differentiates their general solution, using the product rule on product terms, substitutes $x=0$, $\frac{\mathrm{d}y}{\mathrm{d}x}=1$ and solves for the constants. Condone sign slips. |
| | $y=e^{-2x}(2x^2+3x+1)$ | **A1** | oe. |
| | | **[10]** | |
| | **Total for Question 10** | **12** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **11(a)** | $3\cosh x - \sinh x = \frac{3(e^x+e^{-x})}{2} - \frac{e^x-e^{-x}}{2}$ | **M1** | Substitutes exponential definitions of both $\cosh x$ and $\sinh x$. Condone one sign slip. |
| | $= \frac{2e^x+4e^{-x}}{2} = e^x + 2e^{-x}$ | **A1 AG** | Correct substituted expression simplified without error to the printed result; combined steps accepted. |
| | | **[2]** | |
| **11(b)** | $\lambda^2 - \lambda - 2 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it: factorising with product of constants $-2$, formula or completing the square; calculator accepted. |
| | $\lambda = 2,\ -1$<br>$(y =)\, Ae^{2x} + Be^{-x}$ | **A1** | |
| | $(y =)\, ae^x + bxe^{-x}$ | **B1** | Accept with an additional $ce^{-x}$ term. |
| | $y' = ae^x + b(1-x)e^{-x}$, $y'' = ae^x + b(x-2)e^{-x}$<br>$-2ae^x - 3be^{-x} = 6e^x + 12e^{-x}$<br>$\Rightarrow a = \ldots,\ b = \ldots$ | **M1** | Differentiates their particular integral, with an $e^x$ and an $e^{-x}$ term, twice, substitutes into the equation and equates coefficients. Condone differentiation slips. |
| | $a = -3,\ b = -4$ | **A1 A1** | |
| | $y = Ae^{2x} + Be^{-x} - 3e^x - 4xe^{-x}$ | **A1ft** | ft their complementary function plus their particular integral; must be $y =$ a function of $x$. |
| | $1 = A + B - 3$ | **M1** | Substitutes $x = 0$, $y = 1$ into their general solution with two arbitrary constants. |
| | $y' = 2Ae^{2x} - Be^{-x} - 3e^x - 4(1-x)e^{-x}$<br>$1 = 2A - B - 7 \Rightarrow A = \ldots,\ B = \ldots$ | **M1** | Differentiates their general solution, substitutes $x = 0$, $\frac{dy}{dx} = 1$ and solves both equations for $A$ and $B$. Condone slips differentiating $xe^{-x}$. |
| | $y = 4e^{2x} - 3e^x - 4xe^{-x}$ | **A1** | cao; needs $y =$. Condone $+0e^{-x}$ left in. |
| | | **[10]** | |
| | **Total for Question 11** | **12** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **12** | $\lambda^2 - 4\lambda + 4 = 0 \Rightarrow \lambda = \ldots$ | **M1** | Forms the auxiliary equation and solves it by any method. |
| | $\lambda = 2$ (repeated)<br>$(y =)(A + Bx)e^{2x}$ | **A1** | |
| | $y = e^{2x}(ax^3 + bx^2)$ | **B1** | Also accept<br>$e^{2x}(ax^3 + bx^2 + cx + d)$. |
| | $y' = e^{2x}(2ax^3 + (3a + 2b)x^2 + 2bx)$<br>$y'' = e^{2x}(4ax^3 + (12a + 4b)x^2$<br>$+ (6a + 8b)x + 2b)$<br>$\Rightarrow e^{2x}(6ax + 2b) = xe^{2x}$<br>$\Rightarrow 6a = 1,\ 2b = 0$ | **M1 A1** | **M1** Differentiates their trial function of the form $e^{2x} \times$ polynomial twice by the product rule, substitutes and equates coefficients. Condone slips in differentiation.<br>**A1** Correct reduced form, or $6a = 1$, $2b = 0$ seen. |
| | $y = \frac{1}{6}x^3e^{2x}$ | **A1** | |
| | $y = (A + Bx)e^{2x} + \frac{1}{6}x^3e^{2x}$ | **A1ft** | ft their complementary function with two arbitrary constants plus their nonzero particular integral. |
| | $x = 0,\ y = 1 \Rightarrow A = 1$ | **M1** | Substitutes $x = 0$, $y = 1$ into their general solution with two arbitrary constants. |
| | $\frac{\mathrm{d}y}{\mathrm{d}x} = 2e^{2x}\left(A + Bx + \frac{x^3}{6}\right) + e^{2x}\left(B + \frac{x^2}{2}\right)$<br>$0 = 2A + B \Rightarrow B = \ldots$ | **M1** | Differentiates their general solution, product rule on at least one product, substitutes $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ and solves for both constants. |
| | $y = e^{2x}\left(\frac{x^3}{6} - 2x + 1\right)$ | **A1** | oe, with $y =$. |
| | **Total for Question 12** | **10** | |